Chapter II – Data Fundamentals

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Outline

- Types of Data
- Descriptive Statistics
  - Central Tendency
  - Variability
  - Correlation
- Random Variable
- Data Distributions
- Data Cleaning
- Public Datasets

Types of Data

Types of Data

- Major types of Data:
  - Numerical
    - Discrete
    - Continuous



  - Categorical
    - Nominal
      - Do not have an order



  - Ordinal
    - Also categorical but with an intrinsic order



https://unbabel.com/blog/customer-satisfaction-guide/

Numerical / Quantitative Data

- This type of data has mathematical meaning and can be applied to a:
  - Measurement:
    - Person's height, weight, blood pressure;
  - Counter:
    - number of votes, patients, houses …



- Numerical data can be further split into two types:
  - Discrete
    - represent items that can be counted
    - assume possible values that can be listed (finite set of values or may be infinite but countable, from 0 to $\pm\infty$)
  - Continuous
    - Represent measurements
    - Their possible values cannot be counted and can only be described using intervals of real numbers (usually they are rounded off).

Numerical / Quantitative Data

- Numerical data can also be defined as:
  - An **interval-scale** variable is measured on a scale of equally spaced units:
    - Examples:
      - Temperature, when measured in Celsius or Fahrenheit
      - Time, if measured using a 12-hour clock, is an example of interval data.





  - A **ratio-scale** variable is an interval variable with a true zero point (absolute value):
    - Examples:
      - Height in centimeters
      - Duration of illness.



Categorical / Qualitative Data

- Qualitative data has no **inherent mathematical meaning and no intrinsic order**
  - Gender
  - Yes/No; 0/1 binary information **Important in ML!!**
  - Colors, …
  - It is possible to assign numbers to categories in order to represent them more compactly but the numbers don´t have mathematical meaning



Ordinal Data

- Ordinal
  - a mixture of numerical and categorical
    - Numbers assigned to the different categories have some mathematical meaning, as there is a relationship between the different categories
      - e.g. rating a restaurant on a scale from 1 (lowest) to 5 (highest) stars
      - killip scale
      - Satisfaction rate, …
    - Allow analysis based on the median, mode, quantile, percentile, etc.
      - while nominal variables only allow based on mode.
      - the mean might be considered but with some restrictions (https://creativemaths.net/blog/ordinal/)
        - e.g. tripadvisor classification



    - Ordinal variables can be defined based on a numeric value (1, …5) or without a numeric value (very unsatisfied, …., very satisfied)



https://unbabel.com/blog/customer-satisfaction-guide/

Types of Data



Or use QR code

Types of Data

Numerical
Quantitative

Discrete | *Continuous*

Interval Scale

Measurable but arbitrary zero

Ratio Scale

Measurable and absolute (true) zero

Mathematical Meaning

Categorical
Qualitative

Nominal

No rank or ordering
No mathematical meaning

*Ordinal**

* **Intrinsic order**
Natural order or rank
Limited mathematical meaning

Types of Data

- Main Characteristics:

	Nominal	Ordinal	Interval	Ratio
order rank		✓	✓	✓
equal spacing			✓	✓
true absolute zero				✓
add and subtract			✓	✓
mode	✓	✓	✓	✓
median		✓	✓	✓
arithmetic mean			✓	✓

Descriptive Statistics

https://realpython.com/python-statistics/

Descriptive Statistics

- Two main approaches:
  - Quantitative Approach
    - numerical description/summarization of data
  - Graphical
    - Based on charts plots, histograms, …

- Statistics Analysis can also be an:
  - **Univariate analysis** if just the behavior of a single variable is analyzed
  - **Multivariate analysis** concerned with relationships among multiple variables.

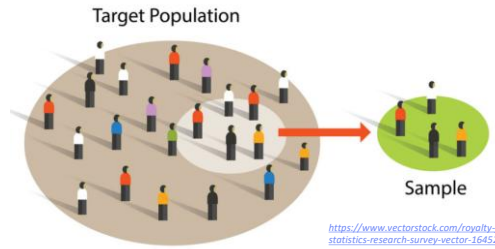
Descriptive Statistics

- Types of Measurements:
  - Central Tendency
    - About the centers of data (e.g. mean, median, mode)
  - Variability
    - About spread of data (e.g. variance, standard deviation)
  - Correlation or Joint Variability
    - About the relation between a pair of variables in a dataset (e.g. covariance, correlation coefficient)

Population/Sample

- **Population** is a set of all elements or items involved in the problem to be solved
  - Problem: impossible to collect and analyze data in all population elements.
    - Solution: To choose a representative subset of that population (sample)

- **Sample**
  - Ideally, the sample should preserve the essential statistical features of the population (represent the **entire population** as accurately as possible).



https://www.vectorstock.com/royalty-free-vector/sample-from-population-statistics-research-survey-vector-16452707

Types of Measures – Central Tendency

Descriptive Statistics

Central Tendency

- The **measurements of central tendency** show the central or middle values of datasets.
  - Mean
  - Weighted mean
  - Median
  - Mode

"A measure of central tendency is a single value that attempts to describe a set of data by identifying the central position within that set of data."

https://statistics.laerd.com/statistical-guides/measures-central-tendency-mean-mode-median.php

Central Tendency

- Sample mean
  - sample arithmetic mean or simply the average, is the arithmetic average of all the items in a dataset.
    - It is mathematically expressed as

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \quad , i = 1, \dots, n$$

  - Arithmetic average as a measure of central tendency is simple and easy to use. However it has important disadvantages/limitations:
    - Sensitive to extreme values
    - Limited to situations when all values are equally important
    - Not suitable for time series type of data

Central Tendency

- Mean
  - Sensitive to extreme values
    - In the presence of extreme values it might be misleading

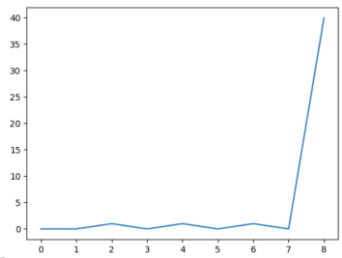
```
import numpy as np
from matplotlib import pyplot as plt

x=np.arange(9)
y=np.array([0, 0, 1, 0, 1, 0, 1, 0, 40])

print('mean:  ', y.mean())
plt.plot(x,y);
```

misleading

```
mean:   4.777777777777778
```



Central Tendency - Mean

- Not suitable for time series type of data (e.g.)
  - A stock investment performed like this in the last two years:
    - Year 1: +70%
    - Year 2: -50%
  - What is the arithmetic return of the investment?
    - $s = \frac{70-50}{2} = 10\%$? (positive return)?
  - Wrong!!
    - Arithmetic mean fails to discover the loss
    - For instance, initial investment of one million dollars, at the end of year 1 the position was worth 1.7 million. In the second year, its value has halved, and ends up at 0.85 million
      - Less than the initial investment. Loss of money!
    - In this case, the simple arithmetic average is **useless and misleading**



Central Tendency

- Sample Weighted Mean
  - weighted arithmetic mean or weighted average, is a generalization of the arithmetic mean that enables to define the relative contribution of each data point to the result.
    - Different $x_i,\ i = 1, \ldots n$ have specific weights $w_i \geq 0$. Then, the weighted mean in given by:

$$\bar{x} = \frac{\sum_i^n w_i x_i}{\sum_i^n w_i}$$

    - Allows to consider the importance of the different contributions

Central Tendency

- Weighted Mean
  - Example: Test grades with different weights

```
test1=np.array([10, 10, 10, 10])  # 20% (4 values)
test2=np.array([14, 14, 14, 14])  # 80% (16 values)

test1_mean=test1.mean()
test2_mean=test2.mean()

tests_means=np.array([test1_mean,test2_mean])

print('mean value:', tests_means.mean())
print('weighted mean value: ', np.average(tests_means, weights=[0.2, 0.8]).round(2))
```

```
mean value: 12.0
weighted mean value:  13.2
```

Central Tendency

- Median
  - It is the middle element of a **sorted dataset**.
    - The dataset can be **sorted** in increasing or decreasing order
      - If the number of elements n of the dataset is odd, then the median is the value at the middle position: $0.5(n + 1)$
      - If n is even, then the median is the arithmetic mean of the two values in the middle, that is, the items at the positions $0.5n$ and $0.5n + 1$.

```python
import numpy as np

nchildren=np.array([2, 1, 0, 2, 0, 1, 2, 1, 9])

print("Mean:", np.mean(nchildren))
print("Median",np.median(nchildren))
print("Sorted array:", np.sort(nchildren))
```

```
Mean: 2.0
Median 1.0
Sorted array: [0 0 1 1 1 2 2 2 9]
```

Median

Central Tendency

- Median
  - Median is less susceptible to the presence of outliers than the mean

```python
nchildren=np.array([2, 1, 0, 2, 0, 1, 2, 1, 30])

print("Mean:", np.mean(nchildren))
print("Median",np.median(nchildren))
```

```
Mean: 4.333333333333333
Median 1.0
```

Influence of outliers in mean *and* median

```python
import numpy as np

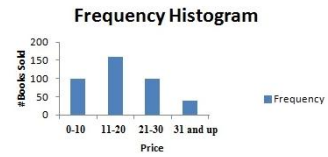
nchildren=np.array([2, 1, 0, 2, 0, 1, 2, 1, 9])

print("Mean:", np.mean(nchildren))
print("Median",np.median(nchildren))
print("Sorted array:", np.sort(nchildren))
```

```
Mean: 2.0
Median 1.0
Sorted array: [0 0 1 1 1 2 2 2 9]
```

Central Tendency

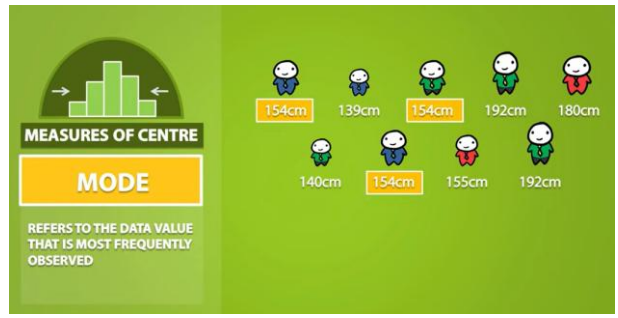
- Median
  - Advantages:
    - It is very simple to understand and easy to calculate.
    - It lies in the middle of the sorted series so it is not affected by the extreme values.
    - It is specially useful in open-ended distributions since the position is what matters



  - Disadvantages:
    - It is less representative than average because it does not depend on all the items in the series.
    - It is not capable of further algebraic treatment.
      - e.g., It is not possible to find a combined median of different groups.

Central Tendency

- Mode
  - the sample mode is the most frequent value in the dataset
    - If there isn't a single value then the set is multimodal since it has multiple modal values.
    - Besides multiple modal issue, mode might not provide a good measure of central tendency as the most common value may be away from the rest of the data in the data set.



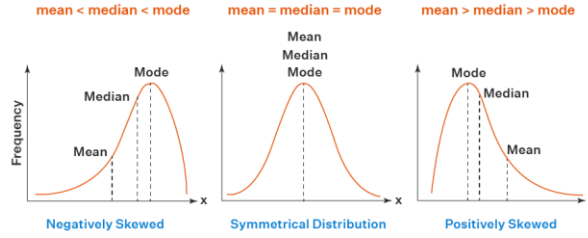
https://www.youtube.com/watch?v=mk8tOD0t8M0

    - Often applied for categorical data where the most common category should be identified

Central Tendency

- When to use mean, median, mode?

Type of Variable	*Central Tendency*
Numerical (not skewed)	Mean
Numerical (skewed)	Median
Categorical	Mode
Ordinal	Median



https://www.cuemath.com/data/relation-between-mean-median-and-mode/

Types of Measures – Measures of Variability

Descriptive Statistics

Measures of Variability

- The measures of central tendency are not sufficient to describe data.
- Measures of variability quantify the spread of data points
  - Variance
  - Standard Deviation
  - Percentiles
  - Skewness / Kurtosis
  - Range

Measures of Variability

- Variance
  - the sample variance quantifies the spread of data, i.e. measures how spread data is.
    - It represents numerically how far the points are from the mean
      - **Population Variance**

$$\sigma^2 = \frac{\sum_{i=1}^{n}(x_i - \mu)^2}{(n)} \quad , where\ i = 1, \ldots, n \quad and \quad \mu\ the\ population\ mean$$

      - **Sample Variance**

$$S^2 = \frac{\sum_{i=1}^{n}(x_i - M)^2}{(n-1)} \quad , where\ i = 1, \ldots, n \quad and \quad M\ the\ sample\ mean$$

Bessel's Correction: https://towardsdatascience.com/the-reasoning-behind-bessels-correction-n-1-eeea25ec9bc9

Measures of Variability

- Variance

```
1 import numpy as np
2 a=np.array([1,4,5,4,8])
3 print("variance", np.var(a))
```

```
variance 5.04
```

- spreading data

```
1 import numpy as np
2 a=np.array([1,4,9,4,12])
3 print("variance", np.var(a,ddof=0))
```

```
variance 15.6
```

Variance increases!!

Note: by default NumPy calculates population variance (ddof=0) to calculate sample variance (ddof=1)

Measures of Variability

- Standard Deviation
  - measure of data spread
  - Sample
    - It is the positive square root of variance
    - It is very often applied **as it has the same unit as the data points**

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - M)^2}{(n-1)}}$$

  - Population standard deviation must consider the population variance (entire population)

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \mu)^2}{n}}$$

Measures of Variability

- Variance/Standard Deviation

Population

$$\sigma^2 = \frac{\sum_{i=1}^{n}(x_i - \mu)^2}{(n)}$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \mu)^2}{n}}$$

Sample

$$S^2 = \frac{\sum_{i=1}^{n}(x_i - M)^2}{(n-1)}$$

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - M)^2}{(n-1)}}$$

$n = 5$

```
a=np.array([1,4,9,4,12])
print('population var', a.var(ddof=0))
print('population std', a.std(ddof=0).round(2))
```

```
population var 15.6
population std 3.95
```

```
print('population var', a.var(ddof=1))
print('population std', a.std(ddof=1).round(2))
```

```
population var 19.5
population std 4.42
```

Influence of sample dimension (n)?

$n = 16$

```
a=np.array([1,4,9,4,12,1,6,7,9,5,4,3,6,7,8,3])
print('population var', a.var(ddof=0).round(2))
print('population std', a.std(ddof=0).round(2))
```

```
population var 8.62
population std 2.94
```

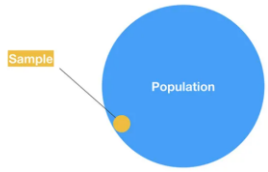
```
print('population var', a.var(ddof=1).round(2))
print('population std', a.std(ddof=1).round(2))
```

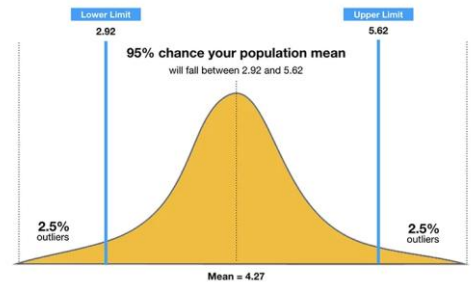
```
population var 9.2
population std 3.03
```

*When **n increases** the difference between sample and population **reduces significantly***

Measures of Variability

- Confidence Interval (CI)
  - Interval that might include a population value with a certain degree of confidence
    - A confidence interval measures **how well your sample represents the population**
      - It represents how much uncertainty there is with any particular statistic.
    - The probability that the confidence interval includes the true statistics value (e.g. mean) within a population is called the confidence level of the CI
      - Most applied is CI 95% (there is a probability of 95% that this interval contains the true value (e.g. population mean)





Measures of Variability

- Sample $\boldsymbol{p}$ Percentile
  - It is the element in the dataset such that $p\%$ of the elements in the dataset are less than or equal to that value.
    - Also, $(100 - p)\%$ of the elements are greater than or equal to that value.
  - Each dataset has three **quartiles**, which are the percentiles that divide the dataset into four parts:
    - The first quartile is the sample 25th percentile.
      - It divides 25% of the smallest items from the rest of the dataset.
    - The second quartile is the sample 50th percentile or the median.
      - 25% of the items lie between the first and second quartiles and another 25% between the second and third quartiles.
    - The third quartile is the sample 75th percentile. It divides 25% of the largest items from the rest of the dataset.

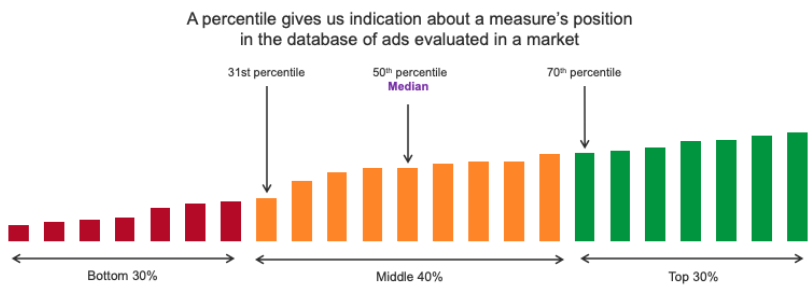
Measures of Variability

- Percentile

```
1 import numpy as np
2 x=np.arange(101)
3 print('percentiles: ',np.percentile(x,[25,50,75]))
```

```
percentiles:  [25. 50. 75.]
```

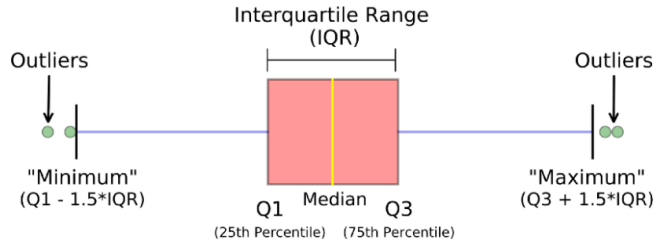
  - meaning? (e.g.)



https://s3-eu-central-1.amazonaws.com/euc-cdn.freshdesk.com/data/helpdesk/attachments/production/77002604860/original/es_jT7e3lb4L0kLLaM-yKaBevRa8G9-f8A.png?1591971955

Measures of Variability

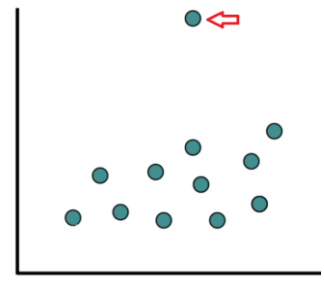
- Median / quartiles
  - Frequently presented as “Box Plot”
    - Very useful for outliers identification



https://towardsdatascience.com/understanding-boxplots-5e2df7bcbd51

Outliers

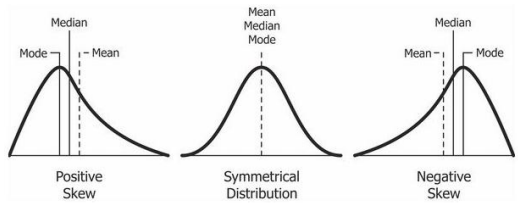
- **Outlier** is a data point that differs significantly from the majority of the data taken from a sample or population
  - Possible causes:
    - Natural variation in data (**even unexpected it is real**)
    - Change in the behavior of the observed system
    - Errors in data collection
  - Errors in data collection
    - Sensor/equipment malfunctioning
    - Data contamination
    - Human error
    - Miscalculations
    - …



Measures of Variability

- Skewness
  - Measures the **asymmetry** of a data sample that is given by $g = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^3/n}{s^3}$

  where $\bar{x}$ is the mean, σ is the population standard deviation, and n is the number of data points.

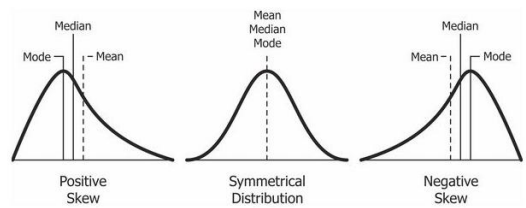


  - **Positive Skewness** means that the tail on the right side of the distribution is longer.
    - The mean and median will be greater than the mode.
  - **Negative Skewness** occurs when the tail of the left side of the distribution is longer than the tail on the right side.
    - The mean and median will be lower than the mode.

Measures of Variability

- Skewness
  - Measures the asymmetry of a data sample

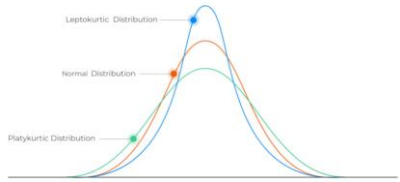
$$g = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^3/n}{s^3}$$



  - g?
    - $-0.5 \leq g \leq 0.5$: the data are fairly symmetrical.
    - $-1 \leq g < -0.5$ (negatively skewed) or $0.5 < g \leq 1$ (positively skewed): the data are moderately skewed.
    - $g < -1$(negatively skewed) or $g > 1$ (positively skewed): the data are highly skewed.
  - Skewness alone doesn't prove normality but might be useful for an initial normality assessment

Measures of Variability

- Kurtosis
  - It is given by: $g_2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^4/n}{s^4}$
    - High kurtosis ($g_2 > 3$) is an indicator that data has heavy tails or lot of outliers.
      - High kurtosis → profusion of outliers → errors, noise, ?
    - Low kurtosis ($g_2 > 3$) is an indicator that data has light tails or lack of outliers.
      - Low kurtosis → Lack of outliers → Check ?



Mesokurtic / Normal Distribution $g_2 = 3$

Leptokurtic: High kurtosis ($g_2 > 3$)

Platykurtic: Low kurtosis ($g_2 < 3$)

  - Similarly to skewness, it also might provide some insight for normality assessment

Measures of Variability

- Range
  - The difference between the maximum and minimum in a dataset

```
1  import numpy as np
2  a=np.array([1,2,5,3,6,7,2,20,1,0,9])
3
4  print('Maximum: ', a.max())
5  print('Minimum: ', a.min())
6
```

```
Maximum:  20
Minimum:  0
```

```
1  print('Range:', a.ptp())
```

```
Range: 20
```

Univariate Analysis

Descriptive Statistics

Univariate Analysis

- A complete univariate analysis focus the exploration of a single variable.
  - Standard sequence of operations:

	Numerical	Categorical
1	Count of: observations unique values	Count of: observations unique values
2	Missing data: Count and Percentage …	Missing data: Count and Percentage …
3	Central tendency: mean median mode	Frequency Analysis: value counts (absolute/relative frequencies)

Univariate Analysis

- A complete univariate analysis focus the exploration of a single variable.
  - Standard sequence of operations:

	Numerical	Categorical
4	Variability (spread): standard deviation variance range interquartile range (IQR) skweness kurtosis	Mode (most frequent category)
5	Outlier Detection: Box Plot IQR	

Univariate Analysis

	Numerical	Categorical
6	Visualizations: histogram box plot violin plot ...	Visualizations: Bar plot Pie chart ...

Types of Measures – Covariance/Correlation

Descriptive Statistics

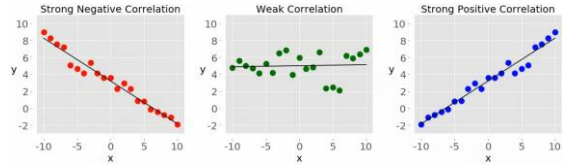
Covariance/Correlation

- Covariance
  - It quantifies the strength and direction of the relationship between a pair of variables x and y
    - It is given by:

$$cov(x, y) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{(n-1)}$$

where $\bar{x}, \bar{y}$ are the respective means and n the number of elements.

- A stronger relationship corresponds to a high value of covariance, it can be:
  - Positive (both variables increase/decrease)
  - Negative (one variable increases/decreases while the other decreases/increases)
- A weak relationship implies a covariance near 0.



https://realpython.com/numpy-scipy-pandas-correlation-python/

Covariance/Correlation

- Covariance
  - Example: Consider 5 observations of 3 variables, x_0, x_1 and x_2 held in the 3 rows of $\mathbf{X}$:

```
X = np.array([ [0.1, 0.3, 0.4, 0.8, 0.9],
               [3.2, 2.4, 2.4, 0.1, 5.5],
               [10., 8.2, 4.3, 2.6, 0.9]
             ])
print( np.cov(X) )
```

The covariance matrix is given by:

```
[[ 0.115   0.0575 -1.2325]
 [ 0.0575  3.757  -0.8775]
 [-1.2325 -0.8775 14.525 ]]
```

The diagonal elements, C_{ii} are the variances in each x_i

  - Some conclusions from the covariance matrix:
    - There is **a strong anti-correlation** between x_0 and x_2 ($C_{02} = -1.2325$) as one increases the other decreases)
    - **No strong correlation** between x_0 and x_1 ($C_{01} = 0.0575$), i.e. x_0 and x_1 do not trend strongly together
  - Problem: How strong is this relationship?
    - Covariance values are hard to interpret, so these values must be normalized (**Correlation Coefficients!**)

Covariance/Correlation

- Correlation Coefficients
  - Provide the relationship between the corresponding elements of two variables in a dataset **in a normalized way.**
  - Very useful to detect multicollinearity situations (e.g. independence of variables in regression)
    - **Positive correlation**: occurs when larger values of x correspond to larger values of y and vice versa

$$r > 0$$

$$r = 1 \quad maximum, perfect\ positive\ linear\ relationship$$

    - **Negative correlation**: occurs when larger values of x correspond to small values of y and vice versa

$$r < 0$$

$$r = -1 \quad minimum, perfect\ negative\ linear\ relationship$$

    - **Weak or no correlation**: when the is no apparent relationship

$$r \approx 0$$

$$weak\ relationship$$

Covariance/Correlation

- Correlation Coefficients
  - Their interpretation depends on the Context / Area of Application:

r	Interpretation
0.7 to 1.0	strong positive
0.5 to 0.7	moderate positive
0.3 to 0.5	weak positive
0.0 to 0.3	negligible/no
-0.3 to 0.0	weak negative
-0.5 to -0.3	moderate negative
-1.0 to -0.7	strong negative

- Social sciences : $r > 0.5$ is often considered strong
- Clinical Area: $r > 0.7$ is typically considered strong.
- Finance: $r > 0.3$ might be considered meaningful.
- Engineering: $r > 0.9$ or higher might be considered strong
- …

Covariance/Correlation

- **Important!**

Correlations **do not not imply causation** only allow to identify relationships in variation

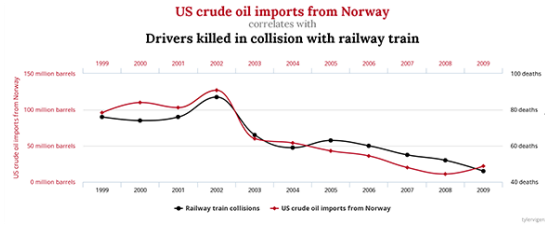
$$r > 0$$
$$r = 1 \quad maximum, perfect\ positive\ linear\ relationship$$

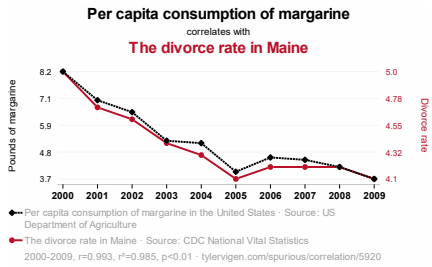
$$r < 0$$
$$r = -1 \quad minimum, perfect\ negative\ linear\ relationship$$

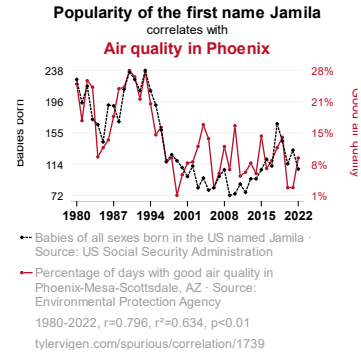
$$r \approx 0$$
$$weak\ relationship$$

Covariance/Correlation

- Spurious Correlations
  - correlation ≠ causation







Covariance/Correlation

- There are 4 types of correlations:

Pearson

Spearman

- Pearson correlation
  - Parametric test, both variables should be **numeric** and **normally** distributed.
- Spearman correlation
  - Does not carry any assumptions about the distribution of the data (**non-parametric test**).
  - It is an appropriate correlation analysis when the variables are measured on a scale that is at least **ordinal**.

Covariance/Correlation

- There are 4 types of correlations:

Kendall rank

Point Bi-Serial

- Kendall rank correlation
  - Also a non-parametric test that can be applied to **ordinal** variables. Usually applied when the sample size is limited.

- Point-Biserial correlation.
  - A correlation measure of the strength of association between a **continuous-level variable** (ratio or interval data) and a **binary variable**.

Covariance/Correlation

- Pearson Correlation Coefficient
  - Linear Correlation between **numeric variables** given by:

Pearson

$$\rho = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x - \bar{x_i})^2} \cdot \sqrt{\sum_{i=1}^{n}(y - \bar{y_i})^2}}$$

where $x_1, \dots, x_n$ and $y_1, \dots, y_n$ are values of both variables and $\bar{x}, \bar{y}$ are the respective means.

- Provides the linear relationship between 2 numeric variables:

sorting y

```
import numpy as np
import scipy.stats
x=np.arange(7)
y=np.array([1,4,2,5,3,5,7])
y_sorted=np.sort(y) # Use np.sort() to return a sorted copy
r,p=scipy.stats.pearsonr(x,y)
print('Coef. Pearson',r.round(2))
r,p=scipy.stats.pearsonr(x,y_sorted)
print('Coef. Pearson y Sorted:',r.round(2))
```

```
Coef. Pearson 0.8
Coef. Pearson y Sorted: 0.99
```

Covariance/Correlation

- Correlation Coefficients
  - Pearson Correlation *pearsonr()*

```
import numpy as np
import scipy.stats
x = np.array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
y=np.array([2, 1, 4, 5, 8, 12, 18, 25, 96, 48])
r, p = scipy.stats.pearsonr(x, y)
print("Pearson Correlation Coefficient:",r)

```

```
Pearson Correlation Coefficient: 0.7586402890911869
```

  - the complete matrix of correlation coefficients can be easily computed *corrcoef()*

```
import numpy as np
import scipy.stats
from matplotlib import pyplot as plt
x = np.array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
y=np.array([2, 1, 4, 5, 8, 12, 18, 25, 96, 48])
print(np.corrcoef(x,y))
```

```
[[1.         0.75864029]
 [0.75864029 1.        ]]
```


Covariance/Correlation

- Point-Biserial correlation.
  - In general, when one of the two variables is categorical a correlation study **cannot** be carried out (does not make sense to detect variations in a categorical variable).
    - In this situation when one of the two variables is categorical a chi-square (X^2) test can be applied
      - **Null Hypothesis:** no relationship exists on the categorical variables in the population so they are independent.

If $p - value < 0.05$ then **null hypothesis is rejected**

[variables are not independent, there is a statistically significant relationship between them]

  - However there is a special case:
    - When the **categorical variable is binary**.



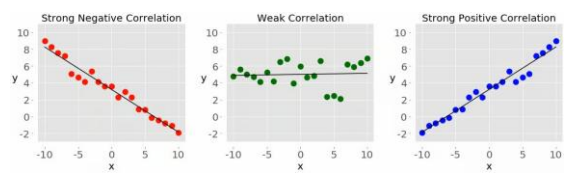
https://www.lifewire.com/bayesian-spam-filtering-1164096

Covariance/Correlation

- When the categorical **variable is binary**.
  - In this case, to calculate the correlation between this variable and the other (numeric) variable, Pearson's correlation coefficient can be applied
    - Point-bisserial correlation coefficient is a special case of Pearson's correlation with the binary variable coded as numeric (0/1)
- The **Point-Biserial Correlation Coefficient** is a correlation measure of the strength of association between a **numeric variable** (ratio or interval data) and a **binary variable**.
- Relationships (binary output):
  - The association between administering the drug (placebo, drug) and the length of survival after treatment
  - Association between gender (female, male) with the earned income
  - …

Covariance/Correlation

- Correlation Coefficient (important!)
  - Bivariate analysis that measures the strength of association between two variables and the direction of the relationship.
    - The value of the correlation coefficient varies between +1 and -1.
    - A value of **± 1 indicates a perfect degree of association between the two variables**. As the correlation coefficient value goes towards 0, the relationship between the two variables will be weaker.
    - The direction of the relationship is indicated by the sign of the coefficient
      - + sign indicates a positive relationship
      - – sign indicates a negative relationship.



Multivariate Analysis

Descriptive Statistics

Univariate Analysis

- A multivariate analysis focus the exploration of the relationships between two or more variables and may involve:

1	Numerical vs. Numerical	Correlation matrix (Pearson, Spearman) Covariance matrix Linear Regression Scatterplots
2	Categorical vs. Categorical	Chi-square test of Independence Two categorical variables are independent or if there's a statistically significant association between them.
3	Numerical vs. Categorical	Group statistics (mean, median, etc. per category) ANOVA (Analysis of variance) Point- Biserial Correlation (only for binary categories) Boxplot (numeric distribution per category)

Univariate Analysis

- Multivariate analysis

4

Visualizations

Pairplot (scatterplots + histograms)
Heatmaps (correlation matrices)
Boxplots
Violin plots
…

Random Variables

Random Variable

- It is a variable whose possible values are numerical outcomes of a random phenomenon:
  - They produce probability distributions based on experimentation, observation, or some other data-generating process.
  - Random variables, allow to understand several real phenomena (real world) based on a sample of data:
    - Knowing the likelihood that a specific value will occur in the real world or at some point in the future.



Random Variable

- Examples:
  - toss of a coin (0/1; head/tail)
  - roll of a dice
  - height of a given person within a population, …
- denoted as X (italic capital letter)
  - e.g. if random variable $X = \{0,1,2,3\}$ then X could be 0, 1, 2 or 3 randomly where each of them might have a different probability.
    - $\neq$ from deterministic variable where the outcome is fixed, i.e. a variable will always assume the same exact value.
  - key element to allow the description of data that are the outcome from a random phenomenon
- Two types of random variables:
  - Discrete: Countable number of distinct values
  - Continuous: Uncountable, takes an infinite number of possible values in some interval

Random Variable

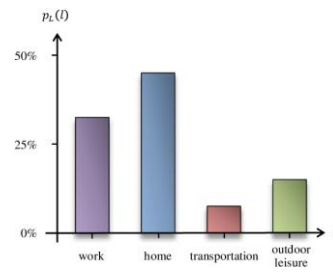
- Some practical examples:

Random Variable	Type
Number of heads obtained in 10 tosses of a fair coin.	
Amount of rainfall in a city on a given day	
Number of customers entering a store between 2 PM and 3 PM.	
Score obtained by a student in a math exam.	
Duration of a person's commute on a weekday.	
Height of a randomly selected adult.	
Number of defective items found in a batch of 100.	
Temperature (in °C or °F) measured at noon each day.	
Number of emails a person receives in a day.	
Time until a light bulb burns out.	

Random Variable

- Discrete Random Variable
  - The probability distribution of a discrete random variable is a list of probabilities associated with each of its possible values.
    - Probability function or the Probability Mass Function (pmf)
      - A discrete random variable X may take k different values (finite or countably infinite), with the probability $P(X = x_i) = p_i$, then pi must satisfy the following:

$$0 < p_i < 1 \;\; ; \;\; for\ each\ i \qquad \sum_{i}^{k} p_i = 1$$



Random Variable

- Continuous Random Variable
  - Takes an infinite number of possible values
    - e.g. A random variable X to be the height of a person. Since the continuous random variable is defined over an interval of values, it is represented by the area under a curve.
  - The Probability Density Function (pdf), takes on continuous values. A curve, which represents a function p(x), must satisfy the following:
    - $p(x) > 0 \quad for\ all\ x$ (no negative values)
    - Probabilities are measured over intervals, **not single points**. That is, the area under the curve between two distinct points defines the probability for that interval

$$p[a \leq x \leq b] = \int_a^b f(x)dx$$

    - The total area under the curve is equal to 1

$$\int_{-\infty}^{+\infty} f(x)dx = 1$$

Random Variable

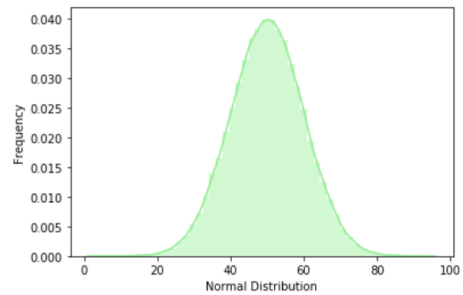
- Continuous Random Variable
  - height of a person randomly selected from a given population

```python
import seaborn as sns
from scipy.stats import norm

data_normal = norm.rvs(size=100000,loc=50,scale=10)
ax = sns.distplot(data_normal,
                  kde=True,
                  color='lightgreen')

ax.set(xlabel='Normal Distribution', ylabel='Frequency')
```



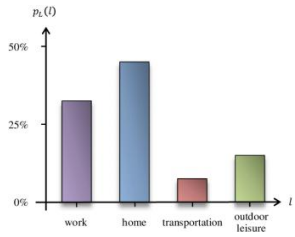


Probability Functions

Probability Mass Function (pmf)

- Discrete data
  - The pmf gives the **probability of discrete values occurring in the dataset**

$$P(X = x_i) = p_i$$

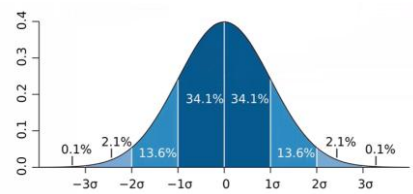


Probability Density Function (pdf)

- Continuous data
  - The probability of a specific value is infinitely small

$$p[a \leq X \leq b] = \int_a^b f_X(x)dx$$

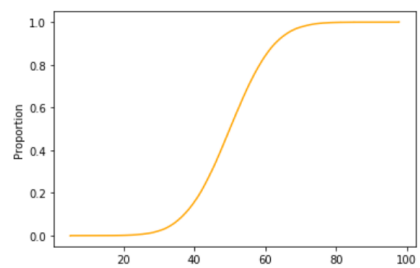
  - The pdf gives the **probability of a data point falling within some given range** of values



Random Variable

- Cumulative distribution function.
  - All random variables (discrete and continuous) have a cumulative distribution function.
    - Gives the probability that the random variable X is less than or equal to x, for every value x.
    - For a discrete random variable, the cumulative distribution function is found by summing up the probabilities.

```
1  import seaborn as sns
2  from scipy.stats import norm
3
4  data_normal = norm.rvs(size=100000,loc=50,scale=10)
5  ax = sns.ecdfplot(data_normal,
6                    color='orange')
7
```



Data Distributions

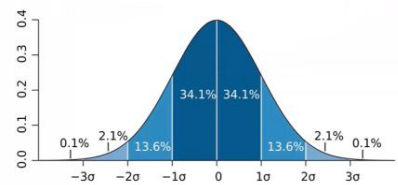
Normal Distribution

Data Distributions

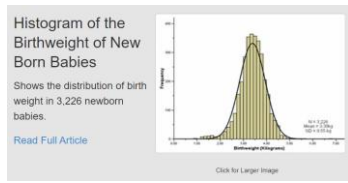
- Data Distribution
  - is a function that provides all the possible values (or intervals) of the data as well as how often each value occurs (probability).
  - Continuous: Some common distributions
    - Normal
    - Uniform*
    - Exponential
  - Discrete:
    - Poisson

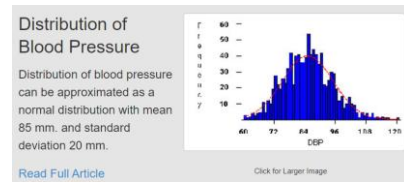
Normal Distribution

- Normal (Gaussian)
  - It is a continuous data distribution



  - It is widely considered, as it tends to represent a large number of data distributions in real life:





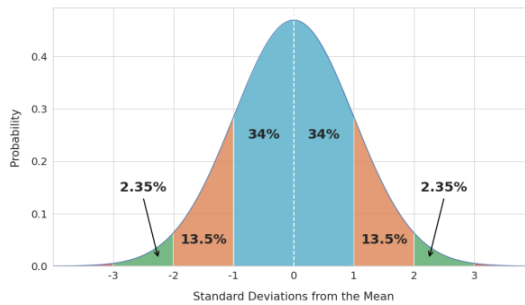
https://galtonboard.com/probabilityexamplesinlife

  - It allows to choose between parametric and non-parametric tests
    - e.g. t-test

Normal Distribution

- Normal (Gaussian)

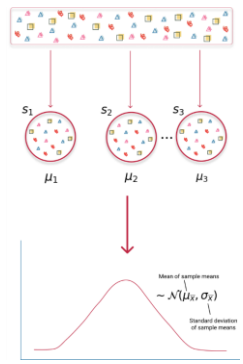
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{(x-\mu)^2}{2\sigma^2}}, \quad \text{where } \sigma>0, \ -\infty < x < \infty$$



  - Symmetric bell shape
  - mean and median are equal (located at the center of the distribution)
  - $\cong 68\%$ of the data falls within 1σ of the mean ($\mu \pm 1\sigma$)
  - $\cong 95\%$ of the data falls within 2σ of the mean ($\mu \pm 2\sigma$)
  - $\cong 99.7\ \%$ of the data falls within 3σ of the mean ($\mu \pm 3\sigma$)

Normal Distribution

- Central Limit Theorem
  - If there is a population with mean μ and standard deviation σ from which sufficiently large random samples (with replacement) are taken, then the distribution **of the sample means** will be approximately normally distributed.
    - This is true regardless of whether the source population is normal or skewed, provided the sample size is sufficiently large (usually $n > 30$).



https://towardsdatascience.com/central-limit-theorem-a-real-life-application-f638657686e1

Normal Distribution

- Confidence Interval (CI)
  - If data follows a **normal distribution**, the confidence level can be calculated using the Normal distribution

$\bar{x}$ - *mean value*
s – standard deviation
n – sample dimension

$$\bar{x} \pm z\frac{s}{\sqrt{n}}$$

Confidence Interval	z
80%	1.282
85%	1.440
90%	1.645
95%	1.960
99%	2.576
99.5%	2.807
99.9%	3.291

  - Otherwise the t-distribution is followed:

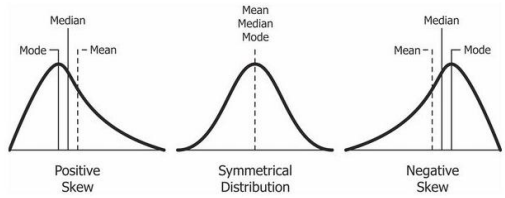
$$\bar{x} \pm t\frac{s}{\sqrt{n}}$$

t ?

One-sided	75%	80%	85%	90%	95%	97.5%	99%	99.5%	99.75%	99.9%	99.95%
Two-sided	50%	60%	70%	80%	90%	95%	98%	99%	99.5%	99.8%	99.9%
1	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	127.3	318.3	636.6
2	0.816	1.080	1.386	1.886	2.920	4.303	6.965	9.925	14.09	22.33	31.60
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	7.453	10.21	12.92
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610

Normal Distribution

- Skewness related with Normal Distribution



  - It is important as it allows to infer about normal distribution:
    - Significant skewness clearly indicates that data are not normal.
      - e.g. some prediction models are heavily based on the unrealistic assumption of normal distribution. These models will provide incorrect estimates (risk, revenues, …).
      - e.g., some classical statistical tests rely on normality assumptions.

Data Distributions

Uniform Distribution
Exponential Distribution
…

Uniform Distribution

- Probability distribution where every possible outcome as an equal likelihood of happening.
  - The probability is constant since each value has equal chance of being the outcome
    - Discrete uniform distributions have a finite number of outcomes.



    - A continuous uniform distribution is a statistical distribution with an infinite number of equally likely measurable values:

$$f_X(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 \ , x < a \vee x > b \end{cases}$$

Some Examples of Uniform Distribution	
Discrete	Probability of each side of a dice
Discrete	Probability of each side of a coin
Continuous	Perfect random number generator
...	....

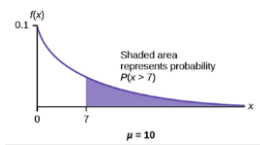
Exponential Distribution

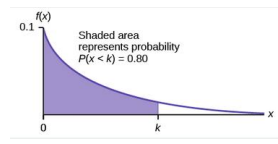
- This distribution is given by the pdf:

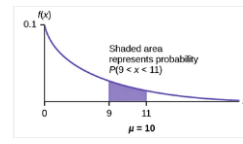
$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0;\ \lambda > 0 \\ 0 & otherwise \end{cases}$$

  - Intends to represent some specific time evolutions:
    - Time will elapse before an earthquake occurs in a given region
    - Time take before a call center receives the next phone call
    - Time a piece of machinery works without breaking down

e.g. Duration of a Computer (years)







$$\lambda = \frac{1}{\mu}$$

https://courses.lumenlearning.com/introstats1/chapter/the-exponential-distribution/

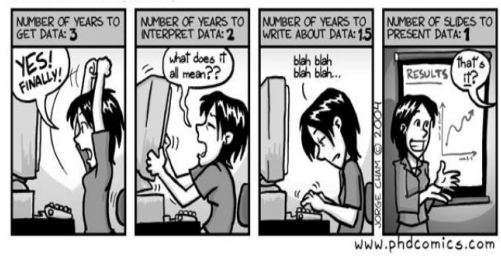
Data Cleaning

Quality of Data

Data Cleaning

"If **80 percent of our work is data preparation**, then ensuring **data quality is the important work** of a machine learning team."

Andrew Ng, founder Coursera, Stanford University



https://phdcomics.com/comics/archive.php?comicid=462

Data Cleaning

- Quality of data?



CORONAVÍRUS

DGS. Bases de dados incompletas e com erros

ATUALIZADO ↗ 4704

Dados da DGS são de fraca qualidade, dizem investigadores: homens grávidos, pessoa com 134 anos, mais de 4.000 dados desaparecidos. DGS diz que dados são de vigilância, não de investigação científica.

https://observador.pt/
10th November 2020

Data Cleaning

*Incorrect or inconsistent data **leads to false conclusions**.*

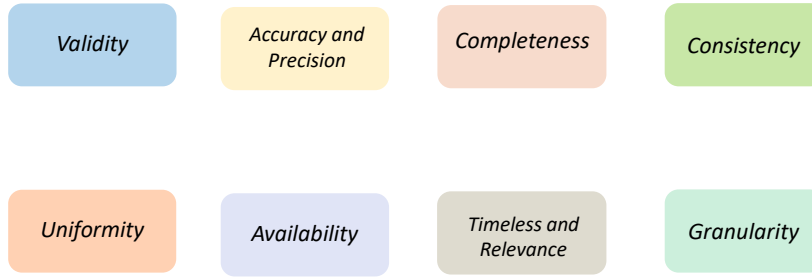
*Data cleaning **has a high impact** on the quality of results.*

A simple algorithm can outweigh a complex one just because it deals with enough and high quality data.

https://towardsdatascience.com/the-ultimate-guide-to-data-cleaning-3969843991d4

Data Cleaning

- Quality of Data



Quality of Data

- Validity
  - The degree to which the data conform to defined business rules or constraints
    - Data Type constraints
      - Values in a particular column must be of a particular type e.g. numeric, boolean, …

	total_bill	tip	sex	smoker
0	16.99	1.01	Female	No
1	10.34	1.66	Male	No
2	21.01	3.50	Male	No
3	23.68	3.31	Male	No
4	24.59	3.61	Female	No

    - Range Constraints
      - Range allowed
    - Set-membership constraints
      - Values of a column must come from a set of discrete values (e.g. person's gender: male/female)

Quality of Data

- Validity
  - Mandatory constraints
    - Some columns cannot be empty.
  - Unique constraints
    - A field must be unique across a dataset
  - Regular expressions patterns
    - Text fields that have to be in certain pattern (e.g. zip code, id card number, …)
  - Cross field validation
    - Some conditions that span across multiple fields must be assured (e.g. patient date of discharge cannot be earlier than the date of admission)

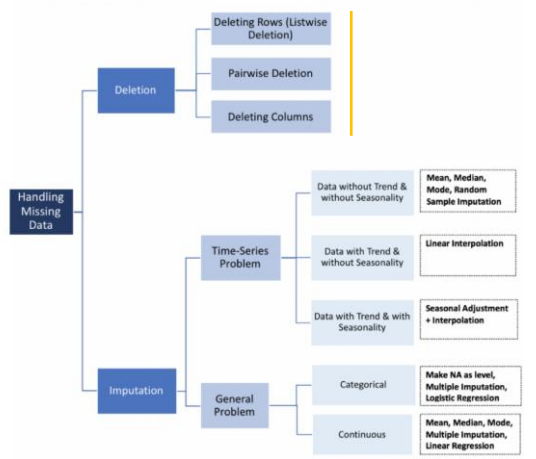
Quality of Data

- Accuracy and Precision
  - The degree to which the data is close to true values
  - This characteristic refers to the exactness of the data.
    - e.g. The temperature of an object is 60 degree Celsius.
      - Thermometer is showing 60 degrees for all readings. That means thermometer is accurate and precise (the optimal condition).



Quality of Data

- Completeness
  - The degree to which all the required data is known. Missing data is a very frequent problem.



To be explored!!

https://towardsdatascience.com/how-to-handle-missing-data-8646b18db0d4

Quality of Data

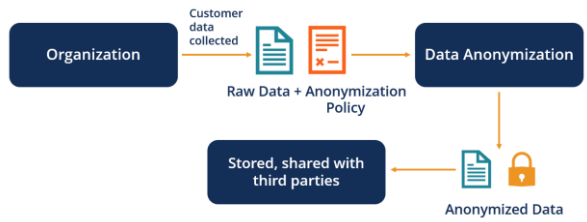
- Consistency
  - The degree to which the data is consistent, within the same data set or across multiple data sets.
    - Examples:
      - A valid age of 10 years cannot match with marital status “divorced”
      - A customer recorded in two tables with different addresses, …
- Uniformity
  - The degree to which the data is specified using the same unit of measure.
  - Data must be converted to a single measure unit
    - Examples:
      - It doesn’t make any sense to use:
        - **different metric system** (e.g. pound, kilos ,…)
        - **different currencies** (EUR (€), USD ($), …)

Quality of Data

- Availability
  - Can be related with legal and regulatory constraints and sometimes can be very difficult to solve (e.g. clinical research).



  - Data anonymization refers to the method of preserving private or confidential information by deleting or encoding identifiers that link individuals to the stored data.
    - To protect the private activity of an individual or a corporation while preserving the credibility of the collected data.



https://corporatefinanceinstitute.com/resources/business-intelligence/data-anonymization/

Quality of Data

- Relevance and Timeliness
  - There must be a valid reason to consider that specific data (relevance)
  - Sometimes data are meaningful if are collected at the right moment in time.
    - Data collected too soon or too late could drive to inaccurate decisions, e.g.:
      - **Climate change research** needs to use recent studies and data, not outdated ones.



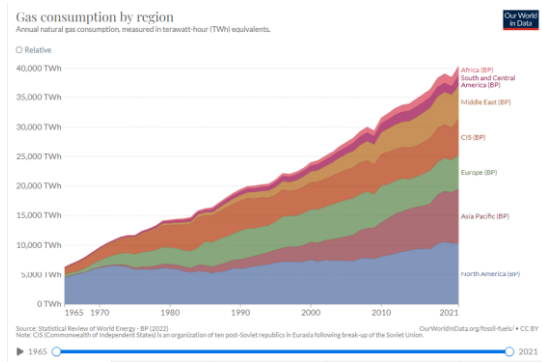
      - **A marketing team needs** to understand current customer preferences and trends to create relevant marketing campaigns.
        - Out-of-date data on customer behavior would make their campaigns useless.



https://www.istockphoto.com/pt/fotos/advertising-campaign

Quality of Data

- Granularity
  - The level of detail at which data is collected must be appropriated to the problem to be solved.
    - An appropriate level of granularity must be assured to provide sufficient uniqueness and distinctive properties to data.



Gas Consumption per Country

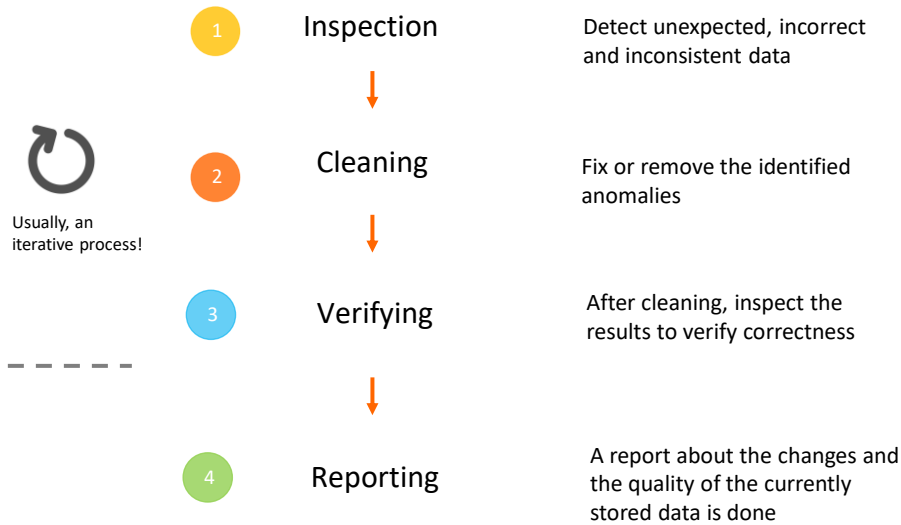
?

https://ourworldindata.org/grapher/natural-gas-consumption-by-region

Data Cleaning

Quality of Data
Workflow

Workflow



Data Cleaning

Quality of Data
Workflow
Inspection

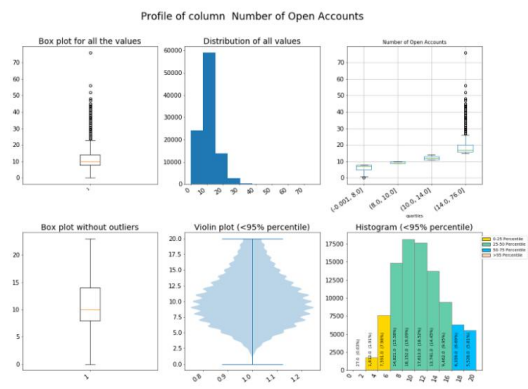
Inspection

- Very important phase in the data cleaning workflow
  - **Data profiling**
    - The process of examining the data available from an existing information source (e.g. a database or a file) and collecting statistics/information about it.
      - Descriptive statistics:
        - e.g. min, max, mean, mode, percentile, standard deviation, frequency, variation, aggregates such as count and sum
      - Metadata information:
        - Data type, dimensions, missing values, patterns, uniqueness,...
      - Time consuming process!

Column 0	age	years_seniority	income	parking_space	attending_party	entree	pets
Tony	48	27		1	5	shrimp	
Donald	67	25	86	10	2	beef	
Henry	69	21	95	6	1	chicken	62
Janet	62	21	110	3	1	beef	
Nick		17		4			
Bruce	37	14	63		1	veggie	
Steve	83		77	7	1	chicken	
Clint	27	9	118	9		shrimp	3

Inspection

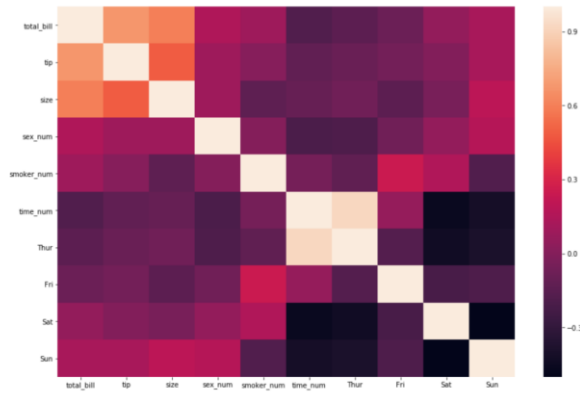
- Visualization
  - e.g.
    - Boxplot
    - Histogram
    - Bar charts
    - Complex visualizations
    - …



https://towardsdatascience.com/automated-data-profiling-99523e51048e

Inspection

- Visualization
  - e.g. correlation plot, very important to identify the potential importance of different features (columns)



Inspection

- Some decision can be taken in this phase:
  - Very important!
    - Any decision regarding the data requires **domain knowledge** or support of someone with **domain knowledge.**
      - Who really understands the data!!
      - Decisions taken **without proper understanding** of the data and the domain knowledge usually result **in poor analysis**.



Inspection: Decisions

Column names and data types	Understand what each column/feature is/represents… The column/feature data type is correct, The columns will be useful for feature engineering …
Numerical statistics (features)	Some errors can be detected
Data Distribution	How to handle the outliers Should the outliers be dropped Which visualization will be effective to identify trends or patterns, etc.
Unique Value Count	Feature type and number of unique values
% or number of null values	Should the instance/row be dropped? Should the column/feature be dropped? Should imputation techniques be applied?
Correlation plot	Could strongly correlated features be replaced by only one without significant loss of information?

Data Cleaning

Quality of Data
Workflow
- Inspection
- **Cleaning**
  - **Irrelevant Data**
  - Duplicates
  - Syntax Errors
  - Scaling
  - Missing Data
  - Outliers
  - Encoding

Cleaning: Irrelevant Data

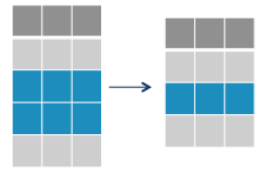
- Different techniques can be applied to remove/correct wrong data
  - Irrelevant Data
    - Actually this data is not important for the domain of a given problem
      - e.g. phone number to analyze the heath status of a population (delete columns)
      - e.g. the goal is to assess the status of population in one country (Portugal) so data from other countries (China) shouldn't be considered. (delete rows)
    - Important!
      - To drop off data the analyst **should be sure that piece of data is irrelevant**, otherwise the respective relevance should be investigated (e.g. correlation)
      - **Domain experts** (e.g. clinical partner, financial consultant, …) should conduct this identification.

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Cleaning: Duplicates

- Duplicates
  - are some instances repeated in the dataset, usually originated if data are combined from multiple sources.



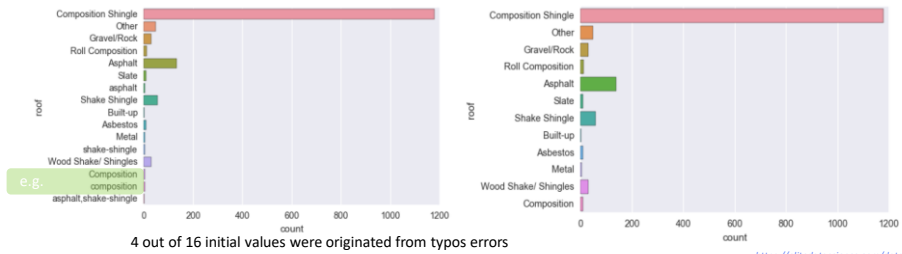
- Type Conversion
  - Numbers stored as numerical data types.
  - Categorical values encoded as numbers
  - Values that can't be converted should be defined as NA (incorrect value that must be fixed)
  - …

Data Cleaning

Quality of Data
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Cleaning: Syntax Errors

- Syntax Errors
  - Fix Typos:
    - Strings can be entered in many different ways frequently with mistakes (e.g. m, Male, M, …; in this all of these values should be replaced by just 1 (e.g. male))
      - Sometimes a bar plot is very useful to detect this situation
    - "not applicable"; "NA", "na", "null", "NULL" mean that the value is missing
    - Consider the existence of extra spaces before and after a string. Should be removed!
    - Some numerical codes can be represented with prepending zeros to ensure they have always the same number of digits (e.g. ID card)

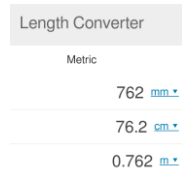


4 out of 16 initial values were originated from typos errors

https://elitedatascience.com/data-cleaning

Cleaning: Standardize

- Standardize
  - Strings
    - All values must be either lowercase or uppercase
  - Numerical
    - All values have a given (the same) unit
      - For a given feature the respective values must adopt the same unit throughout the whole dataset.
        - e.g In the same dataset the animal heights cannot consider different measurement units.

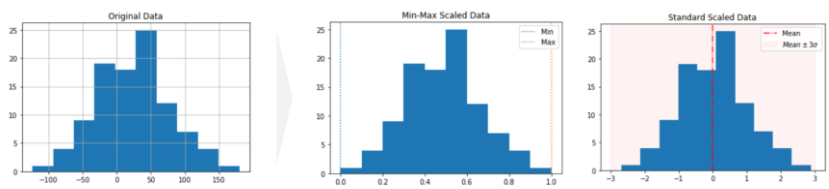


Data Cleaning

Quality of Data
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Cleaning: Scalers

- Scaling (*min-max*)
  - It is the process of converting an actual range of numerical values into a **standard range of values in the interval [0,1]**
  - Some algorithms like KNN, K-means, and SVM are significantly affected by the range of features as they apply distances between data points to determine their similarity.
  - If there are features that have very large values relative to the other features:
    - **These large values can dominate or skew some ML algorithms.**



Cleaning: Scalers

- Scaling (*min-max*)
  - Algorithms may give more weight to the large values and ignore the features with smaller values
    - e.g. data set containing two features:
      - age (ranges from 0–100); income (ranges from 0 –100000)
        - Income is about 1,000 times larger than age. **Very different ranges!**
        - With some algorithms the attribute income will intrinsically influence the result more due to its larger value.
        - However, it doesn't mean that it is more important as a predictor.
    - Other potential advantage of scaling can be related with the speed of learning.

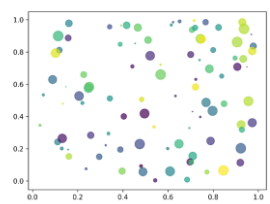
Cleaning: Scalers

- Scaling (*min-max*)
  - The normalization in the interval [0,1] is given by:

$$x_i' = \frac{x_i - \min(x_i)}{\max(x_i) - \min(x_i)}$$

  - To rescale for a given interval $[a, b]$ then following equation must be applied:

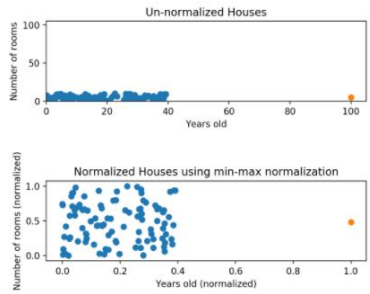
$$x_i' = a + \frac{(x_i - \min(x_i))\ (b - a)}{\max(x_i) - \min(x_i)}$$



https://towardsdatascience.com/everything-you-need-to-know-about-min-max-normalization-in-python-b79592732b79

Cleaning: Scalers

- Scaling (*min-max*)
  - main drawback of this approach: **sensitive to outliers**
    - e.g. Consider 2 features (number of rooms, years old) with different ranges
      - the un-normalized values of "years old" are between 0 and 40 and there is an outlier (100).
      - The normalization min-max will keep this behavior, squishing all* values between 0 and 0.4.



The intervals of "number of rooms" [0,1] and the "years old" [0,0.4] remain different

https://www.codecademy.com/articles/normalization

Cleaning: Scalers

- Normalization (*z-score*) / Standard Scaler
  - z-score normalization is a strategy of normalizing data:

$$\widehat{z_i} = \frac{x_i - \mu}{\sigma}$$

where μ is the mean value of the feature x_i and σ is the standard deviation of the feature x_i.

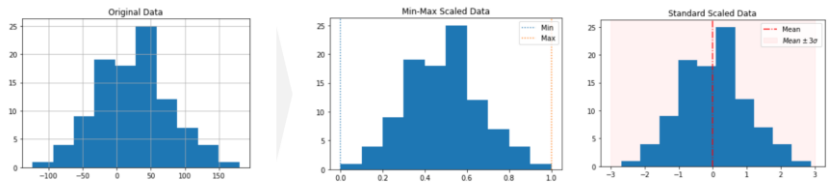
- If x_i is:
  - $x_i = \mu$, it will be normalized to 0.
  - $x_i < \mu$, it will be normalized to a negative number
  - $x_i > \mu$, it will be normalized to a positive number.
- The magnitude of the normalized values (positive/negative) is determined by the standard deviation of the original feature.
  - If the unnormalized data had a large standard deviation then the normalized values will be closer to 0.

Cleaning: Scalers

- Normalization (*z-score*)

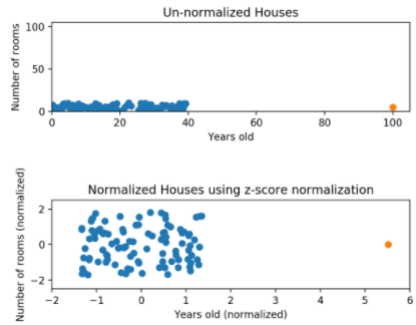
$$\widehat{z_i} = \frac{x_i - \mu}{\sigma}$$

  - It is sensitive to outliers as they influence the calculation of μ and σ
  - When compared with Scaling, Normalization (z-score) is a more radical transformation:
    - **Normalization** changes the range of data and simultaneously centers the data.
    - **Scaling** just changes the range of your data.



Cleaning: Scalers

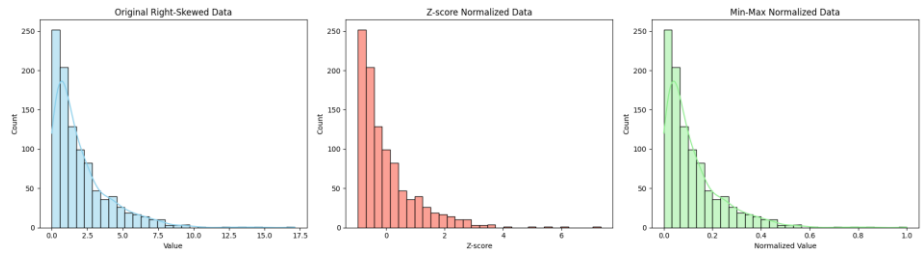
- Normalization (*z-score*)



  - While the data still looks squished, the points are now approximately on the same interval for both features (almost all points rely between -2 and 2 on both axis).

Cleaning: Scalers

- *z-score* / min-max



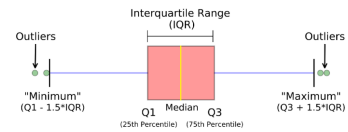
Feature	Z-score Normalization	Min-Max Normalization
Centers data	Yes (mean=0)	No
Rescales data	Yes (std=1)	Yes (range = [0,1])
Affected by skweness	Keeps skew	Keeps skew
Handles outliers	No	No
Distribution changes	No (just centered/scaled)	No (just scaled)

Cleaning: Scalers

- Robust Scaler
  - It is a similar to MinMax but applies the interquartile range (rather the min and max values) which makes it more robust to the presence of outliers.
    - This increased robustness to the presence of outliers is the basis of robust data scaling

$$x_i' = \frac{x_i - median}{Q_3 - Q_1}$$

where Q_1 is the first quartile and Q_3 the third quartile.



  - There are additional techniques to features' scaling
    - https://scikit-learn.org/stable/auto_examples/preprocessing/plot_all_scaling.html
    - https://scikit-learn.org/stable/modules/preprocessing.html#preprocessing-scaler

Data Cleaning

Quality of Data
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Cleaning: Missing Data

- Missing Data
  - Very frequent situation, namely when the data is directly input (e.g. worker, clinician, …)
    - All the values or some measurements are not inserted.



Column 0	age	years_seniority	income	parking_space	attending_party	entree	pets	emergency_contact
Tony	48	27		1	5	shrimp		Pepper
Donald	67	25	86	10	2	beef		Jane
Henry	69	21	95	6	1	chicken	62	Janet
Janet	62	21	110	3	1	beef		Henry
Nick		17		4				
Bruce	37	14	63		1	veggie		NA
Steve	83		77	7	1	chicken		n/a
Clint	27	9	118	9		shrimp	3	None
Wanda	19	7	52	2	2	shrimp		empty
Natasha	26	4	162	5	3			--
Carol		3	127	11	1	veggie	1	****
Mandy	44	2	68	8	1	chicken		null

https://gallery.azure.ai/Experiment/Methods-for-handling-missing-values-1

  - Typical approaches for dealing with missing data:
    - Removing the data with missing features from dataset
    - Apply a data imputation technique.
    - Using a learning algorithm that can deal with missing values (e.g. Naïve Bayes Classifier)

Cleaning: Missing Data

- Missing Data
  - Typical approaches for dealing with missing data:
    - Removing the data with missing features from dataset
      - Possible to apply when number of instances (rows) with missing data is low (< 5%) and the data set is large.
        - In these conditions it is possible to sacrifice some training examples as the loss of information would not be very significant.
      - If a feature has a significant number of missing values (> 80%) then a typical decision is to drop that feature (entire column).
    - Apply a data imputation technique.
    - Using a learning algorithm that can deal with missing values (e.g. Naïve Bayes Classifier)

Cleaning: Missing Data

- Missing Data
  - Imputation: calculate the missing value based on other observations. There are a lot of different techniques:
    - Statistical based on mean or median.

$$\hat{x}^{(j)} \leftarrow \frac{1}{N}\sum_{i=1}^{N} x_i^{(j)} \quad , i = 1, \ldots, N;\ x_i \in \mathbb{R}$$

If there are many missing values both strategies may originate biased data.



	col1	col2	col3	col4	col5
0	2	5.0	3.0	6	NaN
1	9	NaN	9.0	0	7.0
2	19	17.0	NaN	9	NaN

	col1	col2	col3	col4	col5
0	2.0	5.0	3.0	6.0	7.0
1	9.0	11.0	9.0	0.0	7.0
2	19.0	17.0	6.0	9.0	7.0

Mean Imputation

    - This imputation method is easy and fast however it is not very accurate, it does not consider any correlations between features (considers a single column) and CANNOT be applied to categorical features

Cleaning: Missing Data

- Statistical based on Mode:
  - It works with categorical data.
  - It also works based on a single column doesn't taking into account the correlation between features
- Zero or constant imputation
  - Simple but it originates biased data
  - It does not take into account the feature's data distribution

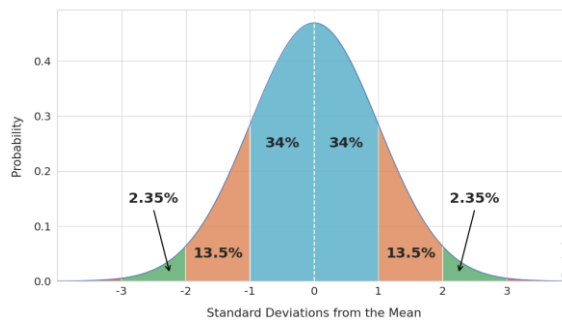
	col1	col2	col3	col4	col5
0	2	5.0	3.0	6	NaN
1	9	NaN	9.0	0	7.0
2	19	17.0	NaN	9	NaN

df.fillna(0)

	col1	col2	col3	col4	col5
0	2	5.0	3.0	6	0.0
1	9	0.0	9.0	0	7.0
2	19	17.0	0.0	9	0.0

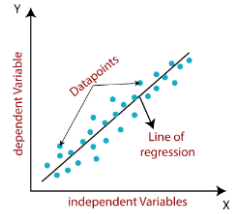
Cleaning: Missing Data

- Statistical based on Normal Distribution
  - When data is normally distributed it can be generated a distribution within 2 standard deviation from the mean (data more likely to be the missing value).
  - The missing values are filled in through the RANDOM generation of numbers of that distribution



Cleaning: Missing Data

- Linear Regression (numerical data)
  - A more advanced technique is to use the missing value as a target variable for a regression problem.



  - The goal is to impute the value of feature x_j in observation i (instance):
    - So in the context of this regression problem $x_j^{(i)}$ assumes the role of target variable $\hat{y}^{(i)}$
  - All the remaining features are applied to form a vector:

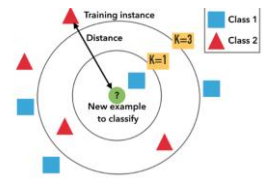
$$\mathbf{x}_m^{(i)} = \left[x_1^{(i)}, x_2^{(i)}, \ldots, x_{j-1}^{(i)}, x_{j+1}^{(i)}, \ldots, x_p^{(i)}\right]$$

$$\hat{y}^{(i)} \leftarrow x_j^{(i)}$$

  - Then a regression model to predict $\hat{y}^{(i)}$ from $\mathbf{x}_m^{(i)}$. Only the examples from original dataset ($\mathbf{x}$,$\hat{y}$) in which the value of x_j is present are applied to train the model.

Cleaning: Missing Data

- Hot-deck imputation
  - Copy values from other similar records. It requires a lot of available data, and some techniques can be applied:
    - Random : the missing value is randomly filled with a value from other instance (observation)
    - Groups: groups are created based on some characteristic (e.g. Male/Female) and the missing value (e.g. height) is filled at random
    - Sequential: The column containing the missing value is sorted according an auxiliary variable so each missing value is filled in with the value of the first following available value.
    - k-Nearest neighbor: the missing value is filled finding the k nearest observations (instances) closest to the instance with missing values
      - Voting, mean, …?



Cleaning: Missing Data

- Other Imputation Methods;
  - Multiple Imputation by Chained Equation (MICE)
  - Deep Learning
    - Datawig https://buildmedia.readthedocs.org/media/pdf/datawig/latest/datawig.pdf
  - …

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Cleaning: Outliers

- Outliers
  - There is not a unique definition for outlier, but it can be defined as:

"any data point that is distinctly different from the rest of your data points. When looking at a variable that is relatively normally distributed, an outliers is any data point that falls 3 or more standard deviations from its mean."



https://www.rapidinsight.com/blog/handle-outliers/

  - Two important aspects:
    - In a large dataset, each individual point carries less weight, so an outlier is less troublesome than the same data point would be in a smaller dataset.
    - The distance to the rest of the dataset is very important.
      - e.g. A point that is ten times as large as the upper boundary will originate more bias than a point that is twice as large.

Cleaning: Outliers

- Outliers belong to one of two categories:
  - Mistake
    - A mistake in the data, could be as simple as typing 10000 rather than 100.00 resulting in a huge impact in the data analysis
  - True outlier
    - it is so different from the other data that can skew the results but it is a true value (e.g. fortune of Bill Gates when compared with the savings of a citizen from US)



Cleaning: Outliers

- Techniques to deal with outliers

Drop Outliers | Limit Outliers | Imputation new value | Transformation

  - Drop the outlier records
    - They should not be removed unless there is a good reason to do that.
    - However in extreme situations (e.g. Bill Gates fortune) it is preferable to remove that record in spite of skewing all the analysis.
      - Very useful when outliers are errors or noise ↑
      - There is the risk of removing valid/meaningful instances ↓
      - If frequently applied can lead to biased models. ↓

Cleaning: Outliers

- Techniques to deal with outliers

Limit Outliers | Imputation new value | Transformation

  - Limit outliers data (Capping / Winsorization)
    - Limit the value of the outlier considering that he behaves in the same way than a lower/higher value. Usually the replacement of extreme values is based on a specified percentile (e.g., 1st or 99th).
      - e.g. might find that people above a certain income level behave in the same way as those with that income
        - This technique preserves all instances ↑
        - It reduces the impact of outliers ↑
        - It can hide valuable information ↓
        - The definition of the threshold is not clear (arbitrary) ↓

Cleaning: Outliers

- Techniques to deal with outliers

Imputation new value

Transformation

- Imputation of a new value
  - If an outlier seems to be due to a mistake an imputation technique (missing data) can be applied. It:
    - maintains the dataset dimensions ↑
    - introduces artificial data ↓
    - increases the risk of bias ↓

Cleaning: Outliers

- Techniques to deal with outliers

Transformation

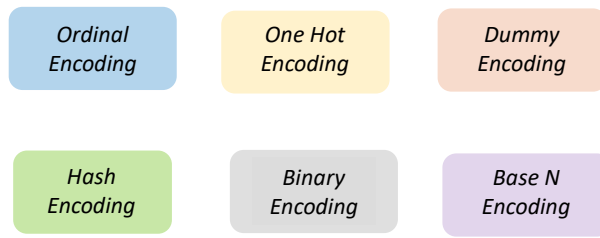
- Transformation
  - In some situations, a transformation (log, percentile, etc.) of the data rather than using the data itself might be considered
    - e.g. create a percentile version of the original values and working with that new values
    - It:
      - maintains all instances (preserves the dataset dimensions) ↑
      - reduces skewness ↑
      - may hinder/distort some relationships between variables ↓
      - may be not effective if outliers are extreme ↓

Data Cleaning

Quality of Data
Workflow
- Inspection
- Cleaning
  - Irrelevant Data
  - Duplicates
  - Syntax Errors
  - Scaling
  - Missing Data
  - Outliers
  - Encoding

Cleaning: Encoding

- Encoding
  - In order to properly feed the supervised algorithms, categorical variables must be converted into numerical values
  - There are several types of Encoding



Cleaning: Encoding

- Label or Ordinal Encoding
  - Applied with ordinal variables (there is an intrinsic order)
    - Should not be applied with nominal variables as **it creates an order/distance** between the several levels that **does not match with reality**

```
encoder= ce.OrdinalEncoder(mapping=[{'col':'Degrees',
                                     'mapping':{'None':0,'High school':1,'Diploma':2,'Bachelors':3,'Masters':4,'phd':5}}],
                           return_df=True)
df_train_transformed = encoder.fit_transform(train_df)
df_train_transformed
```

	Degrees
0	High school
1	Masters
2	Diploma
3	Bachelors
4	Bachelors
5	Masters
6	phd
7	High school
8	High school

	Degrees
0	1
1	4
2	2
3	3
4	3
5	4
6	5
7	1
8	1

Cleaning: Encoding

- One Hot Encoding
  - Applied when the **features are nominal**.
  - In one hot encoding, for each value of a categorical feature, a **new variable is created.**
    - Each category is mapped with a binary variable containing either 0 or 1. Here, 0 represents the absence, and 1 represents the presence of that category.
    - Should be applied only **when categorical feature has few levels, otherwise an explosion of new features** would be originated.

```
encoder=ce.OneHotEncoder(cols='Degrees', handle_unknown='return_nan', return_df=True, use_cat_names=True)
df_train_transformed = encoder.fit_transform(train_df)
df_train_transformed
```

	Degrees
0	High school
1	Masters
2	Diploma
3	Bachelors
4	Bachelors
5	Masters
6	phd
7	High school
8	High school

	Degrees_High school	Degrees_Masters	Degrees_Diploma	Degrees_Bachelors	Degrees_phd
0	1.0	0.0	0.0	0.0	0.0
1	0.0	1.0	0.0	0.0	0.0
2	0.0	0.0	1.0	0.0	0.0
3	0.0	0.0	0.0	1.0	0.0
4	0.0	0.0	0.0	1.0	0.0
5	0.0	1.0	0.0	0.0	0.0
6	0.0	0.0	0.0	0.0	1.0
7	1.0	0.0	0.0	0.0	0.0
8	1.0	0.0	0.0	0.0	0.0

Cleaning: Encoding

- Dummy Encoding
  - Dummy coding scheme is similar to one-hot encoding.
    - It represents a small improvement in relation to one-hot encoding as it origins $N - 1$ new features
    - This encoding is based on a set of binary variables (dummy variables).

```
1 data_encoded=pd.get_dummies(data=train_df,drop_first=True)
2 data_encoded
```

	Degrees
0	High school
1	Masters
2	Diploma
3	Bachelors
4	Bachelors
5	Masters
6	phd
7	High school
8	High school

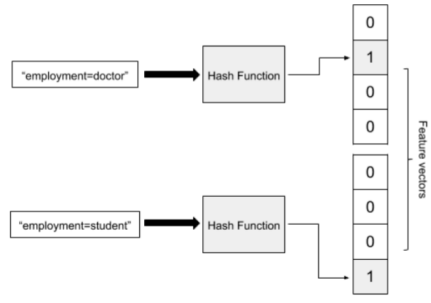
Encoded as: 0000

	Degrees_Diploma	Degrees_High school	Degrees_Masters	Degrees_phd
0	0	1	0	0
1	0	0	1	0
2	1	0	0	0
3	0	0	0	0
4	0	0	0	0
5	0	0	1	0
6	0	0	0	1
7	0	1	0	0
8	0	1	0	0

  - Dummy Encoding as well as One-hot Encoding should not be applied to a categorical variable with high number of levels or when there are several categorical variables throughout the dataset
    - In both situations the number of new features will increase and some unexpected/undesired correlations between the new features may arise.

Cleaning: Encoding

- Hash Encoder
  - The central part of the hashing encoder is the hash function, which maps the value of a category (a key of **arbitrary length; string**) into a **feature vector (predefined length)**.



  - Depending of the number of keys, an hash function may have some collision, i.e. different keys with the same value
    - The degree of collision will depend on the specific hash function.
    - There are some collision resolution strategies.

Cleaning: Encoding

- Hash Encoder
  - This type of encoding is suitable for categorical variables with a large number of levels as it scales extremely well.
    - The user has to specify the number of binary outputs (n-component) of the new feature vector.
    - Since the Hash encoder transforms the data into a fixed n-dimensional vector it may lead to some loss of information

```
encoder=ce.HashingEncoder(cols='Degrees',n_components=4)
df_train_transformed = encoder.fit_transform(train_df)
df_train_transformed
```

	Degrees
0	High school
1	Masters
2	Diploma
3	Bachelors
4	Bachelors
5	Masters
6	phd
7	High school
8	High school

	col_0	col_1	col_2	col_3
0	0	0	1	0
1	0	0	0	1
2	0	0	1	0
3	1	0	0	0
4	1	0	0	0
5	0	0	0	1
6	1	0	0	0
7	0	0	1	0
8	0	0	1	0

Cleaning: Encoding

- Binary Encoder
  - The categorical feature is first converted into numerical using an ordinal encoder, then that number is converted to a binary number, that is split into different columns.
  - should be applied when there are a high number of categories (e.g. cities in a country).

```
encoder=ce.BinaryEncoder(cols='Degrees',return_df=True)
df_train_transformed = encoder.fit_transform(train_df)
df_train_transformed
```

	Degrees
0	High school
1	Masters
2	Diploma
3	Bachelors
4	Bachelors
5	Masters
6	phd
7	High school
8	High school

	Degrees_0	Degrees_1	Degrees_2	Degrees_3
0	0	0	0	1
1	0	0	1	0
2	0	0	1	1
3	0	1	0	0
4	0	1	0	0
5	0	0	1	0
6	0	1	0	1
7	0	0	0	1
8	0	0	0	1

  - Binary encoding is a memory-efficient encoding scheme as it uses fewer features, thus is reduces the curse of dimensionality for data with high cardinality.

Cleaning: Encoding

- Base N Encoding
  - Instead of binary encoding (base 2), it is possible to use a base N encoder.

```
encoder= ce.BaseNEncoder(cols='Degrees', return_df=True,base=5)
df_train_transformed = encoder.fit_transform(train_df)
df_train_transformed
```

	Degrees
0	High school
1	Masters
2	Diploma
3	Bachelors
4	Bachelors
5	Masters
6	phd
7	High school
8	High school

	Degrees_0	Degrees_1
0	0	1
1	0	2
2	0	3
3	0	4
4	0	4
5	0	2
6	1	0
7	0	1
8	0	1

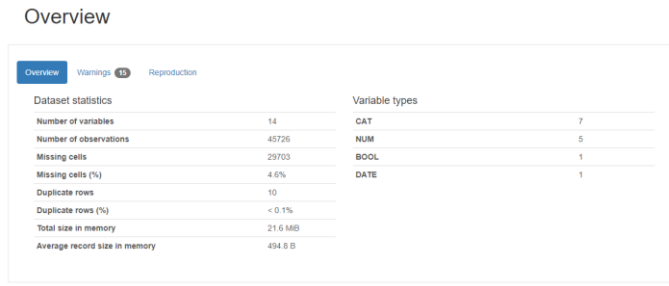
  - BaseN encoding further reduces the number of features required to efficiently represent the data and improving memory usage.
  - There are more encoding techniques that can be applied in the encoding process
    - https://journalofbigdata.springeropen.com/articles/10.1186/s40537-020-00305-w

Data Cleaning

Quality of Data
Workflow
- Inspection
- Cleaning
- **Reporting**

Workflow - Cleaning

- Reporting
  - It should be simple, complete and objective
  - Data Profiling
    - https://towardsdatascience.com/exploratory-data-analysis-8fc1cb20fd15
    - https://pandas-profiling.ydata.ai/examples/master/census/census_report.html

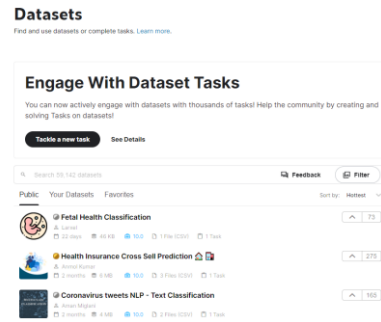


    - https://towardsdatascience.com/automated-data-profiling-99523e51048e
    - https://www.datacamp.com/community/tutorials/python-data-profiling
    - https://towardsdatascience.com/fantastic-pandas-data-frame-report-with-pandas-profiling-a05cde64a0e2

Public Datasets

Datasets

- Some Public/Open datasets:
  - https://www.kaggle.com/datasets
  - https://github.com/awesomedata/awesome-public-datasets
  - https://archive.ics.uci.edu/ml/datasets.php
  - https://registry.opendata.aws/
  - https://www.freecodecamp.org/news/https-medium-freecodecamp-org-best-free-open-data-sources-anyone-can-use-a65b514b0f2d/
  - https://data.europa.eu/euodp/en/home
  - …



https://www.kaggle.com/datasets